

security devices could limit the access to the QR code or Manual Pairing Code to avoid an unauthorized user obtaining the information by simple inspection, or make the QR code and/or Manual Pairing Code removable.

The following is a list of possible ways that are acceptable to satisfy the requirements of inclusion of the QR code, Manual Pairing Code and/or NFC Tag. An entry in the list should not be interpreted as being mutually exclusive with another entry. A device **SHOULD** include as many of these ways as possible.

- QR and Manual Pairing Code shown via an on-device display (when available).
- QR and Manual Pairing Code printed on-device, with removal/obscuring considerations noted above.
- When available, NFC Tag antenna on the device, readable by NFC-enabled Commissioners (e.g., by tapping with a smartphone), with Manual Pairing Code printed on-device, including the removal/obscuring/disabling considerations noted above.
- Manual Pairing Code presented on-device via audio output (when available).
- QR and Manual Pairing Code printed on in-packaging materials.

The following are examples of QR code, Manual Pairing Code and NFC Tag inclusion.

- QR Code and Manual Pairing Code printed on a Matter wireless shade inside the battery compartment cover, and provided in the packaging.
- QR Code and Manual Pairing code on a Matter Smart Thermostat that can be activated via an on-device User Interface and displayed only on screen.
- QR Code and Manual Pairing code for a security sensor that is provided in the packaging, and on-device hidden behind a tamper-monitored cover.
- NFC Tag and Manual Pairing code for a smoke detector present on the back of the device, facing the mounting plate and thus obscured when mounted. Additionally, the mounting plate is metal which prevents NFC tapping when in packaging or mounted.
- QR code provided on an E12 light bulb, with manual pairing code on a removable label (the area of QR code likely fits better on small form factor bulb than the area for a 13 character string).
- A wearable device with only a Manual Pairing Code printed on the fabric. No QR code is present because of the difficulty in scanning a QR code on an irregular surface.
- A Smart speaker, without printed QR or manual pairing code on the device (but possibly in-packaging), that can be triggered to read out a Manual Pairing Code.

5.7.6.1. Inclusion of Onboarding Payload for Concatenated QR Codes

Concatenated QR codes (or NFC Tags) as defined in [Section 5.1.6, “Concatenation”](#) convey the Onboarding Payload for multiple devices provided together (e.g. multi-packs). Therefore, they **SHALL NOT** be used on-device. Rather, they **SHOULD** be provided in packaging. Other requirements and recommendations from the previous section apply, e.g. they **SHALL NOT** be used on the outer packaging.